

Question:
The feedback loop in men which regulates the production of sex hormones includes all but which of the following?
Answer Choices:
A. follicle-stimulating hormone
B. oxytocin
C. pituitary gland
D. hypothalamus
E. gonadotropin-releasing hormone
F. luteinizing hormone
G. testosterone

Reasoning + Answer:

<think>
First , let's understand the feedback loop in men that regulates the production of sex hormones . This loop involves the hypothalamus , pituitary gland , and the testes .

The hypothalamus secretes a hormone called gonadotropin-releasing hormone (GnRH), which stimulates the pituitary gland to release follicle-stimulating hormone (FSH) and luteinizing hormone (LH). FSH and LH then stimulate the testes to produce sex hormones such as testosterone .

The production of sex hormones , specifically testosterone , feeds back to the hypothalamus and pituitary gland , regulating their further release of GnRH , FSH , and LH . Now , let's examine the options :

- A . Follicle-stimulating hormone (FSH) is a part of the feedback loop , as it stimulates the testes to produce sex hormones . So , it is not the correct answer .
- B . Oxytocin is a hormone related to social bonding and childbirth , not directly involved in the sex hormone production feedback loop .
- C . The pituitary gland is indeed a part of the feedback loop , as it receives the GnRH signal from the hypothalamus and releases FSH and LH .
- D . The hypothalamus is the starting point of the feedback loop , secreting GnRH .
- E . Gonadotropin-releasing hormone (GnRH) is the initial hormone that starts the feedback loop by stimulating the pituitary gland to release FSH and LH .
- F . Luteinizing hormone (LH) is a part of the feedback loop , as it stimulates the testes to produce testosterone .
- G . Testosterone is the hormone produced by the testes as a result of the feedback loop , but it is not a regulator of the feedback loop itself .

Considering these roles , the correct answer would be the hormone that is not directly involved in the feedback loop .
</think><answer> B . oxytocin </answer>